

Alexandre Autran

Student in mathematics and machine learning

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Profile

I am a machine learning and mathematics student at the École Normale Supérieure (ENS) in Paris-Saclay and Rennes. My research focuses on probabilistic and geometric methods for machine learning, stochastic processes and stochastic control, high-dimensional statistics, density analysis, optimal transport, graph-based learning, and statistical learning theory.

Education

2024–2027 Normalien - ENS Diploma in Mathematics

Admitted to the ENS through a competitive entrance examination.
Four-year diploma in three years (accelerated track).

2026–2027 Research year in Machine Learning, École Normale Supérieure Paris-Saclay, Paris, France

Final year of the ENS Diploma (research track).

2025–2027 MSc in Mathematics, École Normale Supérieure de Rennes, France

M2 Mathematics. Highest honors (17/20), first semester.
M1 Mathematics. Highest honors (17.6/20). Rank: 5/100+.

2024–2025 BSc in Mathematics, Computer Science minor, École Normale Supérieure de Rennes, France

L3 Mathematics. High honors (15.4/20).

2022–2024 BSc in Mathematics, Computer Science minor, University of Rennes, France

L2 Mathematics. Highest honors (18/20), Rank: 3/100+.
L1 Mathematics. Highest honors (18.5/20). Rank: 1/100+.

Publications

- [PR] [SA] [MS] 1. **A. Autran**, “Mean-Field Fluctuations under Emergent α -Stable Common Noise”, preprint, 2026. [Paper]
- [MF] [OE] [HP] 2. **A. Autran**, “Adaptive Optimal Execution under Locally Stationary Hawkes Order Flow”, preprint, 2026. [Paper]
- [SC] [MF] [FB] 3. **A. Autran**, “Singular Tracking of Fractional Brownian Motion under Proportional Costs”, preprint, 2026. [Paper]
- [PR] [SA] [JD] 4. **A. Autran**, “Improved Wasserstein Estimates for Jump-Diffusion Perturbations under Semi-group Smoothing”, preprint, 2026. [Paper]
- [PR] [SA] [MS] 5. **A. Autran**, “Coupling-Dependent Second-Order Tangents of Emergent α -Stable Common Noise”, preprint, 2026. [Paper]
- [OC] [NA] [MB] 6. **A. Autran**, “Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem”, preprint, 2026. [Paper]
- [SP] [DM] [AM] 7. **A. Autran**, D. Munteanu, and J.-L. Autran, “Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles”, *Applied Sciences*, vol. 16, no. 11, Art. no. 5551, 2026. [Paper]

PR: Probability

OE: Optimal Execution

JD: Jump Diffusions

SP: Semiconductor Physics

SA: Stochastic Analysis

HP: Hawkes Processes

OC: Optimal Control

DM: Device Modeling

MS: Mean-Field Systems

SC: Stochastic Control

NA: Numerical Analysis

AM: Applied Mathematics

MF: Mathematical Finance

FB: Fractional Brownian Motion

MB: Mathematical Biology

Experience

Aug 2026– Machine Learning Researcher

High-Dimensional Density Analysis Using Local Unimodality.

Supervised by Argyris Kalogeratos (Senior Researcher) at Centre Borelli, ENS PS, France.

- May–Aug 2026 **MSc Research Internship**
Optical Modeling of Thermochemical VO₂-Based Oxide Nanocomposites. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Nguyen Ngoc Duy and Amaury Baret (PhD Student) at the University of Liège, Belgium.
- Apr–Jun 2026 **MSc Research Internship**
Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems. [[Report](#)]
Supervised by Prof. Martin J. Gander and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.
- Apr–May 2026 **Machine Learning Projects**
Cardiovascular Disease Prediction Using Statistical Learning and Neural Network Models. [[Report](#)]
Air Pollution Prediction from Environmental Data. [[Report](#)]
Studies carried out as part of the MSc course *Statistical Learning* (Prof. Myriam Maumy).
- Jan 2026 **Statistical Reading Group**
Supervised Statistical Learning – Prediction by Empirical Risk Minimization. [[Presentation](#)]
Supervised by Prof. Magalie Fromont at ENS de Rennes, France.
- Nov–Dec 2025 **MSc Research Seminar**
Point Process and Stein’s Method. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Jean-Christophe Breton at the University of Rennes, France.
- Oct–Nov 2025 **Algebra Reading Group**
Elimination Theory and Extension. [[Presentation](#)]
Supervised by Jade Nardi (Research Scientist) at *ENS de Rennes*, France.
- Jun–Aug 2025 **BSc Research Internship**
Lattice-Based Post-Quantum Cryptography. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Pierre-Alain Fouque and Aurore Guillevic (Research Scientist) at Inria, France.
- May–Jun 2025 **BSc Research Internship**
Algebraic Counting of the Real Roots of Polynomials. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Marie-Françoise Roy at the University of Rennes, France.
- Feb–Mar 2025 **Directed Research Reading**
Geometry of Numbers. [[Report](#)] [[Presentation](#)]
Supervised by Juliette Bavard (Research Scientist) at ENS de Rennes, France.
- Feb–Mar 2024 **BSc Research Internship**
Topological Study of Matrix Groups. [[Report](#)]
Supervised by Prof. Françoise Dal’Bo and Jérémy Bettinger (PhD Student) at the University of Rennes, France.
- May–Jun 2023 **BSc Research Internship**
Introduction to Hyperbolic Geometry and Fuchsian Groups. [[Report](#)] [[Presentation](#)]
Supervised by Frank Loray (CNRS Research Director) at the University of Rennes, France.

Other Activities

- 2024–2025 **ENS Oral Examinations**
Supervision of 2nd year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures: Ulm, Paris-Saclay, Rennes and Lyon. [[Resources](#)]
- 2022–2025 **Teaching in Mathematics**
Tutoring for final-year high school students of the Lycée Chateaubriand in Rennes, and for first and second years BSc students of the University of Rennes. [[Resources](#)]
- 2023–2024 **BSc Student Representative**
Representative on the council of the Mathematics Department of the University of Rennes.
- Before 2022 **High School Competitions**
Concours Général in Mathematics, Physics, Philosophy, and National High School Mathematics Olympiad at Lycée Thiers in Marseille.

Computer Skills

Programming: Python (NumPy, pandas, scikit-learn, PyTorch), C/C++, Java, MATLAB, R.

Languages

French: native language.

English: professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.

Interests: reading and writing, in fantasy and science fiction literature; chess.

Academic Reference

François Bolley [[Web page](#)] Email: francois.bolley@ens-rennes.fr.

Full Professor and Director of the Mathematics Department at ENS de Rennes.